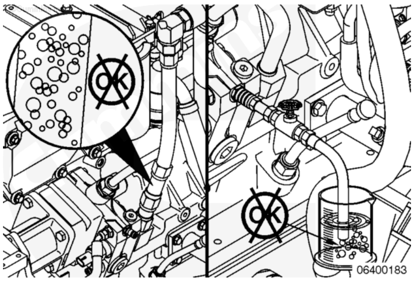
Trainings

General Information

⚠ CAUTION ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



There are two good methods to check for air in the fuel.

- Sight glass method
- Gear pump drain method.

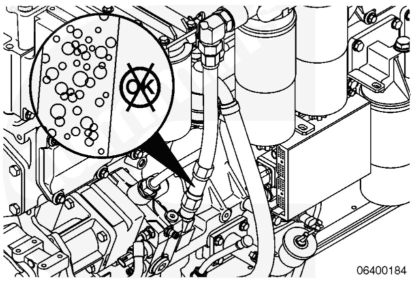
Test

Sight Glass Method

Remove the fuel inlet line connecting the fuel filter head to the fuel pump.

Replace the fuel line with the fuel sight glass, Part Number 3164387, and adapter hose, Part Number 3165146, or equivalent.

Operate the engine at high idle with no load.



Note : A small air leak will have a "milky" appearance.

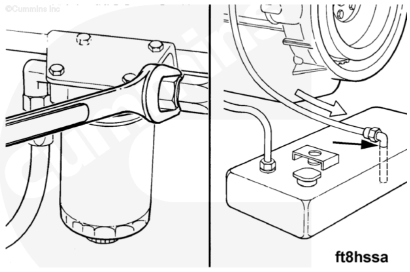
Note : A large air leak will look like bubbles in the fuel.

Note : If the application incorporates a Day tank or a Positive Head Fuel System (PHFS) it is recommended to apply a sight glass before this accessory as these devices may mask the presence of air in fuel upstream.

If an air leak is found, perform the following:

- Systematically inspect the entire fuel supply routing for sources of air ingress starting at the fuel tank followed by all hose/tube interconnections, fuel filtration hardware and day tank/PHFS if provisioned. Tighten any loose connections as needed.

- Check the drop tube in the fuel tank for damage.
- Check the fuel return to tank and ensure the tube is both above fuel level and at a minimum distance of 305 mm [12 in] from the fuel supply connection.
- Check the fuel fitting supply o-rings for damage.



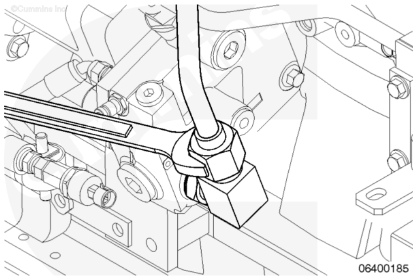
Continue to test and look for a source of air until no air bubbles are visible.

Remove the sight glass.

Tighten the fuel inlet hose.

Torque Value: 88 n•m [65 ft-lb]

Retest engine duplicating the operating conditions when performance complaint occurred to confirm air in fuel correction. This is especially important for standby Generator applications with infrequent starts and may require testing after an extended rest period.

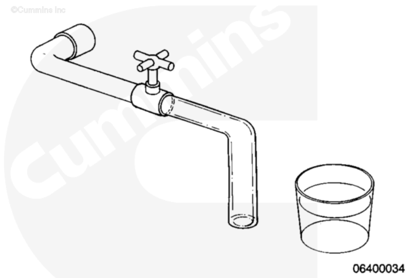


Gear Pump Drain Method

⚠ WARNING ⚠

To perform a pressure side air in fuel test, use the following items:

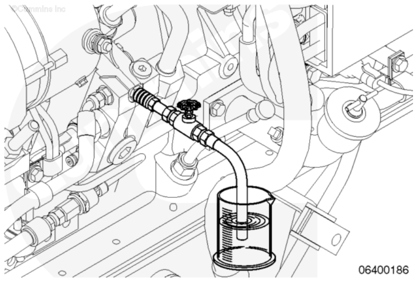
- Quick disconnect fitting, Part Number 3376859
- High pressure hose
- Pressure valve (capable of 2758 kPa [400 psi])
- Clean tubing
- Clean container.



Connect the equipment to the quick-connect fitting at the fuel pump outlet.

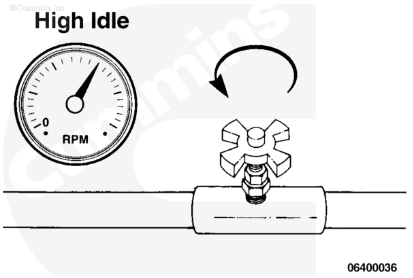
Put the end of the clear hose in the clean container.

Close the pressure valve.



Operate the engine at high idle with no load.

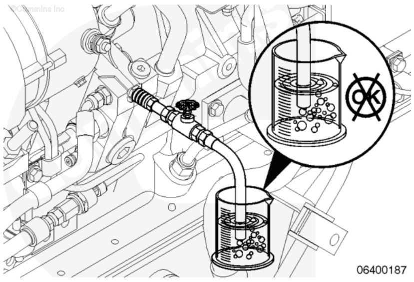
Slowly open the valve until a steady stream of fuel is visible.



Put the end of the hose below the surface of the fuel.

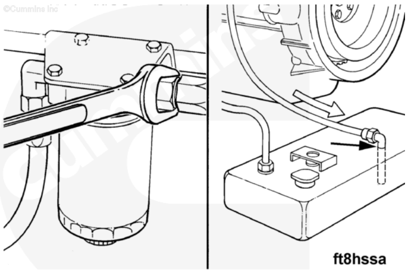
If there is an air leak, bubbles will be visible.

Note : If the application incorporates a Day tank or a Positive Head Fuel System it is recommended to use the Sight Glass Method placed before this accessory as these devices may mask the presence of air in fuel upstream.



If an air leak is found, perform the following:

- Systematically inspect the entire fuel supply routing for sources of air ingress starting at the fuel tank followed by all hose/tube interconnections, fuel filtration hardware and day tank/PHFS if provisioned. Tighten any loose connections as needed.
- Check the drop tube in the fuel tank for damage.
- Check the fuel return to tank and ensure the tube is both above fuel level and at a minimum distance of 305 mm [12 in] from the fuel supply connection.
- Check the fuel supply fitting o-rings for damage.



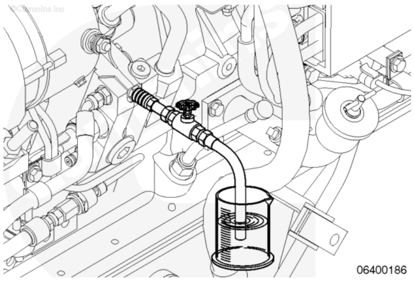
Continue to test and look for air leaks until there are no bubbles visible.

Remove the test equipment.

Tighten the fuel inlet hose.

Torque Value: 120 n•m [89 ft-lb]

Retest engine duplicating the operating conditions when performance complaint occurred to confirm air in fuel correction. This is especially important for standby Generator applications with infrequent starts and may require testing after an extended rest period.



Last Modified: 27-Jan-2023